boolean algebra

mahmood

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Definition 0.0.1. a variable can only be $x \in \{0, 1\}$

1 AND

AND is a binary operation

$$\begin{cases} 0 \cdot 0 = 0 \\ 0 \cdot 1 = 0 \\ 1 \cdot 0 = 0 \\ 1 \cdot 1 = 1 \end{cases}$$

2 OR

OR is a binary operation

$$\begin{cases} 0 + 0 = 0 \\ 0 + 1 = 1 \\ 1 + 0 = 1 \\ 1 + 1 = 1 \end{cases}$$

3 NOT

NOT is a unary operation

$$\begin{cases} 0' = 1 \\ 1' = 0 \end{cases}$$

all of these expressions are equal: $\neg x, \overline{x}, x'$

4 basic properties of boolean algebra

4.1 idempotency

Lemma 1.

$$\begin{cases} x + x = x \\ x \cdot x = x \end{cases}$$

4.2 identity

Lemma 2.

$$\begin{cases} A \cdot 1 = A \\ A + 0 = A \end{cases}$$

4.3 annulment

Lemma 3.

$$\begin{cases} A \cdot 0 = 0 \\ A + 1 = 1 \end{cases}$$

4.4 commutative

Lemma 4.

$$\begin{cases} x + y = y + x \\ x \cdot y = y \cdot x \end{cases}$$

4.5 associativity

Lemma 5.

$$\begin{cases} (x + y) + z = x + (y + z) \\ (x \cdot y) \cdot z = x \cdot (y \cdot z) \end{cases}$$

4.6 complement

Lemma 6.

$$\begin{cases} x + x' = 1 \\ x \cdot x' = 0 \end{cases}$$

4.7 distributivity

Lemma 7.

$$\begin{cases} x \cdot (y + z) = x \cdot y + x \cdot z \\ x + y \cdot z = (x + y) \cdot (x + z) \end{cases}$$

4.8 absorption 1

Lemma 8.

$$\begin{cases} x + x \cdot y = x \\ x \cdot (x + y) = x \end{cases}$$

Proof.

$$\begin{aligned} x + x \cdot y &= x \cdot 1 + x \cdot y \\ &= x(1 + y) \\ &= x \cdot 1 \\ &= x \end{aligned}$$

$$\begin{aligned} x \cdot (x + y) &= (x + 0)(x + y) \\ &= x + 0 \cdot y \\ &= x + 0 \\ &= x \end{aligned}$$

□

4.9 absorption 2

Lemma 9.

$$\begin{cases} x + x' \cdot y = x + y \\ x \cdot (x' + y) = x \cdot y \end{cases}$$

Proof.

$$\begin{aligned} x + x' \cdot y &= (x + x') \cdot (x + y) \\ &= 1 \cdot (x + y) \\ &= x + y \end{aligned}$$

□

4.10 consensus

Lemma 10.

$$\begin{cases} x \cdot y + x' \cdot z + y \cdot z = x \cdot y + x' \cdot z \\ (x + y)(x' + z)(y + z) = (x + y)(x' + z) \end{cases}$$

Proof.

$$\begin{aligned} x \cdot y + x' \cdot z + y \cdot z &= x \cdot y + x' \cdot z + y \cdot z \cdot 1 \\ &= x \cdot y + x' \cdot z + y \cdot z \cdot (x + x') \\ &= xy + x'z + yzx + yzx' \\ &= xy(1 + z) + x'z(1 + y) \\ &= xy + x'z \end{aligned}$$

□

4.11 De Morgan

note

Lemma 11. not sure yet what the 4th formula refers to

$$\begin{cases} \overline{(\overline{x})} = x \\ \overline{(x + y)} = \overline{x} \cdot \overline{y} \\ \overline{(xy)} = \overline{x} + \overline{y} \\ [T(x_1, x_2, \cdots, x_n, 0, 1, +, \bullet)]' = T(x_1', x_2', \cdots, x_n', 1, 0, \bullet, +) \end{cases}$$

5 switching expression

Definition 5.0.1. a **switching expression** is a finite combination of switching variables and the operations AND,OR,NOT

Characteristic 1. a combination of switching expressions results in a switching expression, so if T_1 and T_2 are both switching expressions then $T_1' + T_2$ is also a switching expression

Example 5.0.1. an example of simplifying a switching expression

$$\begin{aligned}
T(x,y,z) &= x'y'z + yz + xz \\
&= z(x'y' + y + x) && \text{distributive property} \\
&= z(x' + y + x) && \text{absorption 2 property} \\
&= z(1 + y) && \text{complement property} \\
&= z \cdot 1 && \text{annulment property} \\
&= z && \text{identity property}
\end{aligned}$$

Example 5.0.2.

$$\begin{aligned}
T(x,y,z) &= (x+y)[x'(y'+z')]' + x'y' + x'z' \\
&= (x+y)[x + (y'+z')'] + x'y' + x'z' \\
&= (x+y)[x + yz] + x'y' + x'z' \\
&= x + yz + x'y' + x'z' \\
&= x + yz + y' + x'z' \\
&= x + yz + y' + z' \\
&= x + z + y' + z' \\
&= x + 1 + y' \\
&= 1
\end{aligned}$$

6 switching function

a function $f(x_1, x_2, \cdots, x_n)$ that outputs only 1 or 0 for any combination of variables it receives
f can only output 2^n different values

Example 6.0.1.

$x_1, \cdots, x_n$	$f(x_1, \cdots, x_n)$	f_1	f_2
$0, \cdots, 0$	$f(0, \cdots, 0)$	1	0
$0, \cdots, 1$	$f(0, \cdots, 1)$	0	0
$0, \cdots, 10$	$f(0, \cdots, 10)$	0	1
$1, 1, 1, \cdots, 10$	$f(1, 1, 1, \cdots, 0)$	0	1
$1, 1, 1, \cdots, 1$	$f(1, 1, 1, \cdots, 1)$	1	1

the number of possible functions for n variables is 2^{2^n}
the complement of f, written as f', is the function that outputs the value 1 wherever f outputs 0
$f + g$ is the sum of both functions g and f, $f + g = 1$ if and only if atleast one of both functions gives 1 for a given combination
$f \cdot g$ is the multiplication of both functinos, for a given combination, it gives 1 if and only if both functions give 1 for that specific combination

Example 6.0.2.

x_1	x_2	x_3	f	f'	g	$f+g$	$f \cdot g$
0	0	0	0	1	1	1	0
0	0	1	1	0	1	1	1
0	1	0	1	0	0	1	0
0	1	1	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	1	0	0	0
1	1	0	1	0	0	1	0
1	1	1	0	1	1	1	0

given f and g are switching functions with n variables, we say f covers g and write $g \subseteq f$ if

$$g(x_1, \cdots, x_n) = 1 \implies f(x_1, \cdots, x_n) = 1$$

if $g \subseteq f$ and g is a multiplication of variables then g is called an of f and we write $g \to f$

Lemma 12. $g \subseteq h + g$

Proof.

$$g = 1 \Rightarrow g + h = 1$$
$$g \cdot h = 1 \Rightarrow g = 1$$

□

Lemma 13. if $g \subseteq f$ and $f \subseteq g$ then $f = g$

Proof.

$$g = 1 \Leftrightarrow f = 1$$

□

Lemma 14. $g + h \subseteq f$ if and only if $g \subseteq f$ and $h \subseteq f$

Proof. 1st side, assume $g \subseteq f$ and $h \subseteq f$, we need to prove $g + h \subseteq f$
assume that $g + h = 1$, then $g = 1$ is a consequence of $g \subseteq f$ that $f = 1$, and $h = 1$ is a consequence of $h \subseteq f$ so that $f = 1$, and so $g + h \subseteq f$
2nd side, assume that $g + h \subseteq f$
if $g = 1$ then $g + h = 1$ and so $f = 1$ and so $g \subseteq f$
if $h = 1$ then $g + h = 1$ and so $f = 1$ and so $h \subseteq f$

□

6.1 minterm

Definition 6.1.1. an expression that is a series of multiplications that contain the variables $x_1, \cdots, x_n$ of a function $f(x_1, x_2, \cdots, x_n)$ is called a **minterm**

6.2 maxterm

Definition 6.2.1. an expression that is a series of additions that that contain the variables $x_1, \cdots, x_n$ of a function $f(x_1, x_2, \cdots, x_n)$ is called a **maxterm**

6.3 DNF

Definition 6.3.1. when a function is written as a sum of products we get what is called as **Disjunctive Normal Form**

6.4 CNF

Definition 6.4.1. when a function is written as a product of sums we get what is called as **Conjunctive Normal Form**

6.5 finding the switching expressions of a switching function

we write a truth table for the function and for each row we write a minterm or a maxterm depending on whether the function outputs 1 or 0, respectively

	x	y	z	f	minterms	maxterms
0	0	0	0	1	$x'y'z'$	
1	0	0	1	0		$x+y+z'$
2	0	1	0	1	$x'yz'$	
3	0	1	1	1	$x'yz$	
4	1	0	0	0		$x'+y+z$
5	1	0	1	0		$x'+y+z'$
6	1	1	0	1	xyz'	
7	1	1	1	1	xyz	

$f(x,y,z) = \sum(0,2,3,6,7) = x'y'z' +$
$x'yz' + x'yz' + xyz' + xyz \longrightarrow$ sum of products, DNF
$f(x,y,z) = \Pi(1,4,5) = (x+y+z')(x'+y+z)(x'+y+z') \longrightarrow$ product of sums, CNF

6.6 prime implicant

a multiplication expression p is called a **Prime implicant** of a switching function f if:

1. $p \to f$ (f covers p or p is an implicant of f)

2. deletion of every literal in p results in product that isnt covered by f meaning $p_1, NOT \to f$ (there is atleast one combination where $p_1 = 0$ but $f = 0$)

in the last example $h = x'y$ is a prime implicant of f because x' and y are implicants of f, k isnt a prime implicant of f
the removal of z, $x'yz$ turns into $x'y$ which is still an implicant of f
the removal of $y \to x'z$ implicant of f
the removal of $x' \to yz$ implicant of f

Lemma 15. every sum of products that isnt minimizable that equals f is a sum of prime implicants of f

Proof. assume in contradiction that there is a sum of products that isnt minimizable that equals f and contains the product φ that isnt a prime implicant of f
assume $f = g + \varphi$, in case of the sum where $\varphi = 1$ and $f = 1$ and so $\varphi \to f$, in case that φ isnt a prime implicant then its possible to remove from φ atleast 1 literal so that we get φ_0 so that $\varphi_0 \to f$
also $\varphi \to \varphi_0$ in case φ is a product of literals that φ_0 is a part of
.....

□

6.7 essential prime implicant

from previous theorems

1. a minimal switching expression of a function is the sum of its *Prime Implicants*

2. **Prime Implicants** correspond to squares in the Karnaugh map that arent a part of bigger squares

<u>question</u>: which *Prime Implicants* exist in the minimal switching expression
<u>definition</u>: a product of literals P is called an *Essential Prime Implicant* of the function f if:

1. P is a *Prime Implicant* of f

2. P covers atleast a minterm of f that cant be covered by another *Prime Implicant*

3. <u>note</u>: a minimal switching expression has to contain all the possible *EPI's* (*Essential Prime Implicants*)

6.8 the process of finding a minimal switching expression of a switching function

given the switching function f

- find all the prime implicants of f

- from the list of prime implicants, find the essential prime implicants and put them in an expression

- remove from the list of PI all the EPI's and all the PI's that are covered by EPI's

- if the set of EPI's covers f, then we're done, otherwise, choose more PI's so that f is entirely covered by the resulting set of PI's but make sure to pick a minimal number of PI's to do the job, and make sure to pick the PI's with the least number of literals

> **note**
> this isnt a definite solution for the problem but its enough for functions with a small number of literals

using karnaugh maps we can apply this process
each cell maps to a combination of variables, we place each minterm in its corresponding cell, i.e. with the cell that provides the correct corresponding values for the variables

$b \backslash a$	0	$\overline{a}$ 1
0	0	2
b | 1	1	3

$c \backslash ab$	00	01	a 11	10
0	0	2	6	4
c | 1	1	3	7	5

b

$cd \backslash ab$	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

by finding rectangles whose area is a power of 2 in these tables we would be finding minterms and the least amount of rectangles we can find would give us the minimal form we're looking for
the variables that cells in a rectangle may share are the ones to be taken for the switching expression

Example 6.8.1.

$$f(w,x,y,z) = \sum(4,5,8,12,13,14,15)$$
$$f = xy' + wx + wy'z'$$

$yz \backslash wx$	00	01	11	10
00	0	1	1	1
01	0	1	1	0
11	0	0	1	0
10	0	0	1	0

Figure 1: *Prime implicants*: $w\bar{y}\bar{z}, wx, x\bar{y}$, all of them are *EPI*

Example 6.8.2.

$$f(x,y,z) = \sum(0,2,3,4,5,7)$$

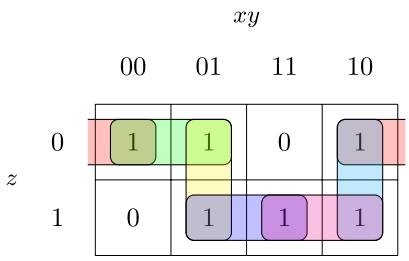

Figure 2: *Prime Implicants*: $\bar{x}\bar{z}, \bar{x}y, yz, xz, x\bar{y}, \bar{y}\bar{z}$

none of these *PI*'s isnt *EPI* and all of them are the same size
this function has 2 minimal forms
$f(x, y, z) = \bar{x}y + xz + \bar{y}\bar{z}$
$f(x, y, z) = \bar{x}\bar{z} + yz + x\bar{y}$

6.9 dont care

dont care is a value of a function we get for a combination of values for the variables that isnt defined, and therefore can be either 1 or 0
the symbol of the value *Don't care* is ϕ, its allowed to switch the value ϕ with either 1 or 0 according to our needs, this is useful when finding the minimal form of a switching function

6.10 map expressions

Definition 6.10.1. a map expression is a switching expression that appears in the map and that isnt 0, 1 or ϕ

Characteristic 2. the usage of map exxpressions allows us to describe a function with more than n variables using a karnaugh map with n variables

Example 6.10.1. consider the following function $f(x, y, z, A, B, C)$

x	y	z	A	B	C	f
0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	0	1	0	0
0	0	0	0	1	1	0
0	0	0	1	0	0	0
0	0	0	1	0	1	0
0	0	0	1	1	0	0
0	0	0	1	1	1	0
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0	1	1	0	1	0	1
0	1	1	0	1	1	1
0	1	1	1	0	0	1
0	1	1	1	0	1	1
0	1	1	1	1	0	1
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	0	0	0	1	1
1	0	0	0	1	0	0
1	0	0	0	1	1	1
1	0	0	1	0	0	0
1	0	0	1	0	1	1
1	0	0	1	1	0	0
1	0	0	1	1	1	1
1	0	1	0	0	0	1
1	0	1	0	0	1	1
1	0	1	0	1	0	0
1	0	1	0	1	1	0
1	0	1	1	0	0	1
1	0	1	1	0	1	1
1	0	1	1	1	0	0
1	0	1	1	1	1	0
1	1	0	0	0	0	0

we can represent it using a 3 variable karnaugh

map

$$ABC + ABC + ABC + ABC = B(AC + AC + AC + AC) = B$$

		wx			
		00	01	11	10
yz	0	0	A	A	C
	1	B	1	ϕ	B'

7 functional completeness

Definition 7.0.1. a set of operations forms a **functionally complete** system if every switching function can be described as a switching expression that only contains operations from that set

Example 7.0.1. for example $\{\neg, +\}$ is functionally complete because $x \cdot y = (x' + y')'$ and in every switching expression where we have $x \cdot y$ we can replace it with $\neg(\neg x + \neg y)$

Example 7.0.2. $\{\cdot, \neg\}$ is also functionally complete

Lemma 16. a functionally complete system consisting of only 1 function returns the complement of a variable if its passed to all its inputs, and the opposite is true, as in a system that doesnt abide by this rule isnt functionally complete

Example 7.0.3. given the completely functional system $\{f(x, y, z, w)\}$, according to this rule, $f(x, x, x, x) = \overline{x}$

Example 7.0.4. take the functionally complete operation NOR, we pass x to all its inputs and we get $\overline{x}$:

$$x \downarrow x = \overline{x} \cdot \overline{x} = \overline{x}$$

Lemma 17. one way to prove that a set of operations is completely functional is by describing all the operations of a set that is already known to be functionally complete using the operations in that set

Example 7.0.5. proof of NOR's functional completeness using the already functionally complete set $\{\neg, +\}$:

$$x \downarrow x = x' \cdot x' = x'$$
$$(x \downarrow y) \downarrow (x \downarrow y) = (x' \cdot y')' \cdot (x' \cdot y')' = (x+y)(x+y) = (x+y)$$
$$((x+y)' + (x+y)')' = (x'y' + x'y')' = (x'y')' \cdot (x'y')' = (x+y)(x+y) = (x+y)$$

7.1 list of systems known to be functionally complete

- $\{+, \cdot, \neg\}$
- $\{+, \neg\}$
- $\{\cdot, \neg\}$, because $x + y = \overline{(\bar{x} \cdot \bar{y})}$
- *NOR*: $x \downarrow y = \overline{x} \cdot \overline{y} = \overline{(x+y)}$
- *NAND*: $x \uparrow y = \overline{x} + \overline{y} = \overline{(xy)}$

7.2 partial functional completeness

Definition 7.2.1. a system of operations is **partially functionally complete** if it need to be combined with 0 and 1 or both to provide functional completeness

note

even if a set of operations can produce 1's and 0's, its still gonna be partially complete if we substitute 1 or 0 to provide completeness

Example 7.2.1.

$$f_1(x,y) = x \oplus y \qquad f_2(x,y) = x + y'$$

$f_1(x,y) = x'y + xy'$
$f_1(x,1) = x' \longrightarrow$ derived negation
$f_2(x, f_2(0,y)) = x + y \longrightarrow$ derived OR operator
so this system is partially complete

8 Shannon's law

Definition 8.0.1. for every switching function $f(x_1, \cdots, x_n)$ we get:

$$f(x_1, \cdots, x_n) = x_1 \cdot f(1, x_2, \cdots, x_n) + x_1' \cdots f(0, x_2, \cdots, x_n)$$
$$f(x_1, \cdots, x_n) = [x_1 + f(0, x_2, \cdots, x_n)] \cdot [x_1' + f(1, x_2, \cdots, x_n)]$$

Characteristic 3. this law helps us move between DNF and CNF forms without looking at the truth table of a function

Example 8.0.1.

$$\begin{aligned} f(x,y,z) &= y'xz + y'z'x' + xz \\ &= x(y'1z + y'z'1' + 1z) + x'(y'0z + y'z'0' + 0z) \\ &= x(y'z + y'z'0 + z) + x'(0 + y'z' + 0) \\ &= x(y'z + z) + x'(y'z') \end{aligned}$$

9 XOR

x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic 4.

$$A \oplus B = B \oplus A \qquad \text{commutative} \tag{1}$$
$$(A \oplus B) \oplus C = A \oplus (B \oplus C) \qquad \text{associative} \tag{2}$$
$$A \bullet (B \oplus C) = (AB) \oplus (AC) \qquad \text{distributive} \tag{3}$$
$$A \oplus \bar{A} = 1,\ A \oplus 1 = \bar{A},\ A \oplus 0 = A \tag{4}$$
$$\overline{A \oplus B} = 1 \text{ only if } A = B \tag{5}$$

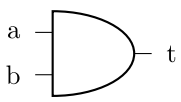
Characteristic 5. if $A \oplus B = C$ then:

$$A \oplus C = A \oplus (A \oplus B) = (A \oplus A) \oplus B = B$$
$$B \oplus C = B \oplus (A \oplus B) = (B \oplus B) \oplus A = A$$
$$A \oplus B \oplus C = A \oplus B \oplus (A \oplus B) = 0$$

10 logic gate

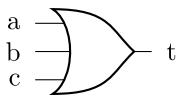
10.1 AND gate

$$t = a \bullet b$$



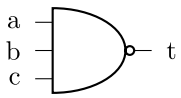
10.2 OR gate

$$t = a + b + c$$



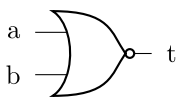
10.3 NAND gate

$$t = \overline{(a \bullet b \bullet c)}$$



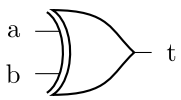
10.4 NOR gate

$$t = \overline{(a + b)}$$



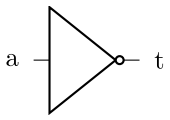
10.5 XOR gate

$$t = a \oplus b$$



10.6 NOT gate

$$b = \overline{a}$$



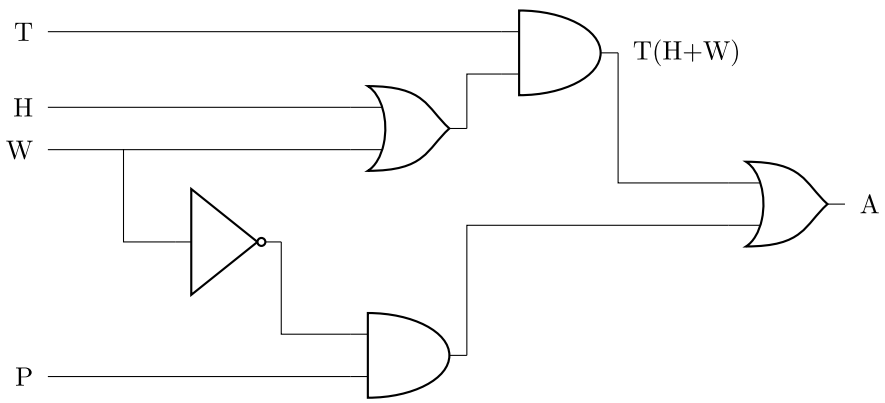
11 logic circuit

a combination of logical gates to represent a more complex switching function is called a logical circuit or a switching circuit

the variables of the function are called the *inputs* of the circuit and the output of the function is the *exit* of the circuit

the exit of a logical circuit depends only on the current combination of values of its input

$$A = T \bullet (H + W) + \overline{W} \bullet P$$



11.1 Digital abstraction

how do we represent the values 0 and 1 using a digital logic circuits?

on one side the logic circuits have only 2 variables, voltage and time and they can have any value in a given range

and on the other side, boolean algebra has only 2 values which are the constants 0 and 1

the solution:

- voltage represents logical value, for example, a high voltage represents 1 and a low voltage represents 0, other voltages, for example a voltage between "high" and "low" are defined as not represnting of logical values

- sometimes, it is agreed upon that the output of a circuit represents the value of the switching function, other times its agreed upon that it doesnt

11.1.1 Logical levels

the two logical values 0 and 1 are represented in *Electronic circuits* using logical levels.

an electronic circuit receives electrical energy from a power supply like a battery, the power supply is the high voltage *Vplus* and the low voltage is called *Vminus*

the levels of voltages denoted v in a circuit is limited in the range $Vminus \leq v \leq Vplus$, the amplitude of voltages is defined as $\Delta v = Vplus - Vminus$

as an example:

$$Vplus - 0.2\Delta v = 4volt \leq v \leq 5volt = Vplus \qquad \text{represents logical 1}$$
$$Vminus = 0volt \leq v \leq 1volt = Vminus + 0.2\Delta v \qquad \text{represents logical 0}$$